15. Examine which open [respectively, closed] intervals are mapped by $f(x) := x^2$ for $x \in \mathbb{R}$ onto open [respectively, closed] intervals.

16. Examine the mapping of open [respectively, closed] intervals under the functions $g(x) := 1/(x^2+1)$ and $h(x) := x^3$ for $x \in \mathbb{R}$.

17. If $f : [0,1] \to \mathbb{R}$ is continuous and has only rational [respectively, irrational] values, must f be constant? Prove your assertion.

18. Let $I := [a,b]$ and let $f : I \to \mathbb{R}$ be a (not necessarily continuous) function with the property that for every $x \in I$, the function f is bounded on a neighborhood $V_{\delta_x}(x)$ of x (in the sense of Definition 4.2.1). Prove that f is bounded on I.

19. Let $J := (a,b)$ and let $g : J \to \mathbb{R}$ be a continuous function with the property that for every $x \in J$, the function g is bounded on a neighborhood $V_{\delta_x}(x)$ of x. Show by example that g is not necessarily bounded on J.

Section 5.4 Uniform Continuity

Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. Definition 5.1.1 states that the following statements are equivalent:

(i) f is continuous at every point $u \in A$;

(ii) given $\varepsilon > 0$ and $u \in A$, there is a $\delta(\varepsilon, u) > 0$ such that for all x such that $x \in A$ and $|x-u| < \delta(\varepsilon,u)$, then $|f(x)-f(u)| < \varepsilon$.

The point we wish to emphasize here is that δ depends, in general, on *both* $\varepsilon > 0$ and $u \in A$. The fact that δ depends on u is a reflection of the fact that the function f may change its values rapidly near certain points and slowly near other points. [For example, consider $f(x) := \sin(1/x)$ for $x > 0$; see Figure 4.1.3.]

Now it often happens that the function f is such that the number δ can be chosen to be independent of the point $u \in A$ and to depend only on ε. For example, if $f(x) := 2x$ for all $x \in \mathbb{R}$, then

$$|f(x)-f(u)| = 2|x-u|,$$

and so we can choose $\delta(\varepsilon,u) := \varepsilon/2$ for all $\varepsilon > 0$ and all $u \in \mathbb{R}$. (Why?)

On the other hand if $g(x) := 1/x$ for $x \in A := \{x \in \mathbb{R} : x > 0\}$, then

$$g(x) - g(u) = \frac{u-x}{ux}. \tag{1}$$

If $u \in A$ is given and if we take

$$\delta(\varepsilon,u) := \inf\left\{\tfrac{1}{2}u, \tfrac{1}{2}u^2\varepsilon\right\}, \tag{2}$$

then if $|x-u| < \delta(\varepsilon,u)$, we have $|x-u| < \frac{1}{2}u$ so that $\frac{1}{2}u < x < \frac{3}{2}u$, whence it follows that $1/x < 2/u$. Thus, if $|x-u| < \frac{1}{2}u$, the equality (1) yields the inequality

$$|g(x)-g(u)| \leq (2/u^2)|x-u|. \tag{3}$$

Consequently, if $|x-u| < \delta(\varepsilon,u)$, then (2) and (3) imply that

$$|g(x)-g(u)| < (2/u^2)\left(\tfrac{1}{2}u^2\varepsilon\right) = \varepsilon.$$

We have seen that the selection of $\delta(\varepsilon,u)$ by the formula (2) "works" in the sense that it enables us to give a value of δ that will ensure that $|g(x)-g(u)| < \varepsilon$ when $|x-u| < \delta$ and

$x, u \in A$. We note that the value of $\delta(\varepsilon, u)$ given in (2) certainly depends on the point $u \in A$. If we wish to consider *all* $u \in A$, formula (2) does not lead to one value $\delta(\varepsilon) > 0$ that will "work" simultaneously for all $u > 0$, since $\inf\{\delta(\varepsilon, u) : u > 0\} = 0$.

In fact, there is no way of choosing one value of δ that will "work" for all $u > 0$ for the function $g(x) = 1/x$. The situation is exhibited graphically in Figures 5.4.1 and 5.4.2 where, for a given ε-neighborhood $V_\varepsilon(\frac{1}{2})$ about $\frac{1}{2} = f(2)$ and $V_\varepsilon(2)$ about $2 = f(\frac{1}{2})$, the corresponding maximum values of δ are seen to be considerably different. As u tends to 0, the permissible values of δ tend to 0.

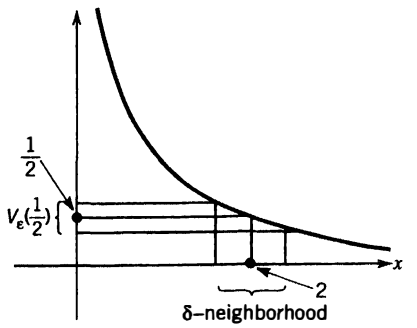


Figure 5.4.1 $g(x) = 1/x \quad (x > 0)$

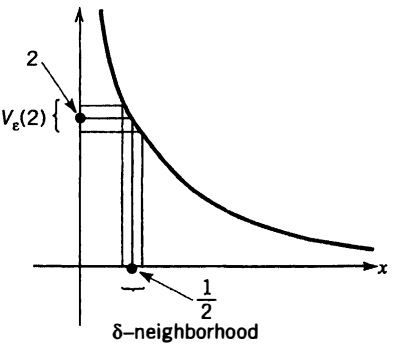


Figure 5.4.2 $g(x) = 1/x \quad (x > 0)$

5.4.1 Definition Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. We say that f is **uniformly continuous** on A if for each $\varepsilon > 0$ there is a $\delta(\varepsilon) > 0$ such that if $x, u \in A$ are any numbers satisfying $|x - u| < \delta(\varepsilon)$, then $|f(x) - f(u)| < \varepsilon$.

It is clear that if f is uniformly continuous on A, then it is continuous at every point of A. In general, however, the converse does not hold, as is shown by the function $g(x) = 1/x$ on the set $A := \{x \in \mathbb{R} : x > 0\}$.

It is useful to formulate a condition equivalent to saying that f is *not* uniformly continuous on A. We give such criteria in the next result, leaving the proof to the reader as an exercise.

5.4.2 Nonuniform Continuity Criteria *Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. Then the following statements are equivalent:*

(i) *f is not uniformly continuous on A.*

(ii) *There exists an $\varepsilon_0 > 0$ such that for every $\delta > 0$ there are points x_δ, u_δ in A such that $|x_\delta - u_\delta| < \delta$ and $|f(x_\delta) - f(u_\delta)| \geq \varepsilon_0$.*

(iii) *There exists an $\varepsilon_0 > 0$ and two sequences (x_n) and (u_n) in A such that $\lim(x_n - u_n) = 0$ and $|f(x_n) - f(u_n)| \geq \varepsilon_0$ for all $n \in \mathbb{N}$.*

We can apply this result to show that $g(x) := 1/x$ is not uniformly continuous on $A := \{x \in \mathbb{R} : x > 0\}$. For, if $x_n := 1/n$ and $u_n := 1/(n+1)$, then we have $\lim(x_n - u_n) = 0$, but $|g(x_n) - g(u_n)| = 1$ for all $n \in \mathbb{N}$.

We now present an important result that assures that a continuous function on a closed bounded interval I is uniformly continuous on I. Other proofs of this theorem are given in Sections 5.5 and 11.3.

5.4.3 Uniform Continuity Theorem *Let I be a closed bounded interval and let $f : I \to \mathbb{R}$ be continuous on I. Then f is uniformly continuous on I.*

Proof. If f is not uniformly continuous on I then, by the preceding result, there exists $\varepsilon_0 > 0$ and two sequences (x_n) and (u_n) in I such that $|x_n - u_n| < 1/n$ and $|f(x_n) - f(u_n)| \geq \varepsilon_0$ for all $n \in \mathbb{N}$. Since I is bounded, the sequence (x_n) is bounded; by the Bolzano-Weierstrass Theorem 3.4.8 there is a subsequence (x_{n_k}) of (x_n) that converges to an element z. Since I is closed, the limit z belongs to I, by Theorem 3.2.6. It is clear that the corresponding subsequence (u_{n_k}) also converges to z, since

$$|u_{n_k} - z| \leq |u_{n_k} - x_{n_k}| + |x_{n_k} - z|.$$

Now if f is continuous at the point z, then both of the sequences $(f(x_{n_k}))$ and $(f(u_{n_k}))$ must converge to $f(z)$. But this is not possible since

$$|f(x_n) - f(u_n)| \geq \varepsilon_0$$

for all $n \in \mathbb{N}$. Thus the hypothesis that f is not uniformly continuous on the closed bounded interval I implies that f is not continuous at some point $z \in I$. Consequently, if f is continuous at every point of I, then f is uniformly continuous on I. Q.E.D.

Lipschitz Functions

If a uniformly continuous function is given on a set that is not a closed bounded interval, then it is sometimes difficult to establish its uniform continuity. However, there is a condition that frequently occurs that is sufficient to guarantee uniform continuity. It is named after Rudolf Lipschitz (1832–1903) who was a student of Dirichlet and who worked extensively in differential equations and Riemannian geometry.

5.4.4 Definition Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. If there exists a constant $K > 0$ such that

$$|f(x) - f(u)| \leq K|x - u| \tag{4}$$

for all $x, u \in A$, then f is said to be a **Lipschitz function** (or to satisfy a **Lipschitz condition**) on A.

The condition (4) that a function $f : I \to \mathbb{R}$ on an interval I is a Lipschitz function can be interpreted geometrically as follows. If we write the condition as

$$\left|\frac{f(x) - f(u)}{x - u}\right| \leq K, \qquad x, u \in I, x \neq u,$$

then the quantity inside the absolute values is the slope of a line segment joining the points $(x, f(x))$ and $(u, f(u))$. Thus a function f satisfies a Lipschitz condition if and only if the slopes of all line segments joining two points on the graph of $y = f(x)$ over I are bounded by some number K.

5.4.5 Theorem *If $f : A \to \mathbb{R}$ is a Lipschitz function, then f is uniformly continuous on A.*

Proof. If condition (4) is satisfied, then given $\varepsilon > 0$, we can take $\delta := \varepsilon/K$. If $x, u \in A$ satisfy $|x - u| < \delta$, then

$$|f(x) - f(u)| < K \cdot \frac{\varepsilon}{K} = \varepsilon.$$

Therefore f is uniformly continuous on A. Q.E.D.